



TRIOS CYBER

LABORATORY PRACTICE 18

Wazuh Component Health Check and Post-Installation Verification

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1. Laboratory Objective and Post-Installation Context

After the Wazuh all-in-one deployment was completed, a component health check was performed on the Ubuntu Server virtual machine. The validation focused on the requirements of Day 18: confirm that the Wazuh server components were running, verify that the Wazuh Dashboard remained accessible from the Windows host, and document the troubleshooting actions that were required to obtain a stable installation. The server hostname used for the deployment was trioscyber and remote administration was performed from Windows PowerShell through the existing VirtualBox NAT port-forwarding configuration.

2. Remote Access to the Wazuh Server

The Ubuntu Server was reached from the Windows host using SSH through host TCP port 2222. The remote session opened successfully with the matthew account and displayed the Ubuntu 24.04.4 LTS server environment. The system summary showed the server online with address 10.0.2.15 on interface enp0s3, confirming that the administrator could reach the Wazuh host before performing the service checks.

```
PS C:\WINDOWS\system32> ssh -p 2222 matthew@127.0.0.1
matthew@127.0.0.1's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of dom 20 sep 2026 02:32:13 UTC

System load:  3.51               Processes:           155
Usage of /:   25.1% of 96.56GB   Users logged in:    0
Memory usage: 4%                IPv4 address for enp0s3: 10.0.2.15
Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

https://ubuntu.com/workshop

El mantenimiento de seguridad expandido para Applications está desactivado

Se pueden aplicar 147 actualizaciones de forma inmediata.
78 de estas son actualizaciones de seguridad estándares.
Para ver estas actualizaciones adicionales, ejecute: apt list --upgradable

Active ESM Apps para recibir futuras actualizaciones de seguridad adicionales.
Vea https://ubuntu.com/esm o ejecute «sudo pro status»

Last login: Sat Sep 19 16:19:11 2026 from 10.0.2.2
matthew@trioscyber:~$
```

3. Core Wazuh Component Health Check

The four components used by the all-in-one deployment were checked with `systemctl is-active`. The Wazuh manager, Wazuh indexer, Wazuh dashboard, and Filebeat each returned active. This confirms that the server-side analysis service, indexing layer, web interface, and event-forwarding component were all running at the time of the Day 18 validation.

```

matthew@trioscyber:~$ echo "==== WAZUH MANAGER ===="
sudo systemctl is-active wazuh-manager

echo "==== WAZUH INDEXER ===="
sudo systemctl is-active wazuh-indexer

echo "==== WAZUH DASHBOARD ===="
sudo systemctl is-active wazuh-dashboard

echo "==== FILEBEAT ===="
sudo systemctl is-active filebeat
==== WAZUH MANAGER ====
active
==== WAZUH INDEXER ====
active
==== WAZUH DASHBOARD ====
active
==== FILEBEAT ====
active

```

Component	Verification	Observed status
Wazuh manager	systemctl is-active wazuh-manager	active
Wazuh indexer	systemctl is-active wazuh-indexer	active
Wazuh dashboard	systemctl is-active wazuh-dashboard	active
Filebeat	systemctl is-active filebeat	active

4. SSH Tunnel and Dashboard Reachability

Because the Wazuh server remained behind the VirtualBox NAT network, browser access was provided through an SSH local-port-forwarding tunnel. Windows TCP port 8443 was mapped through the SSH session to 127.0.0.1:443 on the Ubuntu Server. The tunnel was established with the same forwarded SSH connection on host port 2222 and was kept open while the web interface was tested.

```

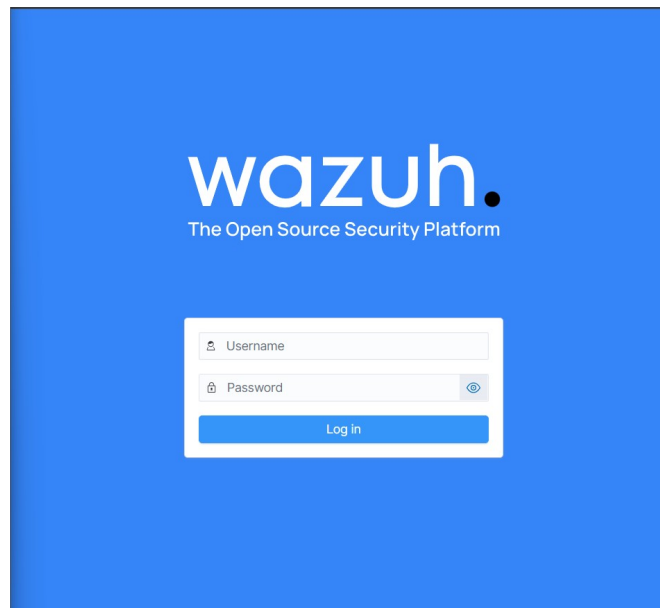
PS C:\Users\User> ssh -N -L 8443:127.0.0.1:443 -p 2222 matthew@127.0.0.1
matthew@127.0.0.1's password:

```

Access element	Observed value
Windows browser endpoint	https://127.0.0.1:8443
SSH connection	matthew@127.0.0.1 using host port 2222
Forwarded destination	127.0.0.1:443 on the Wazuh server

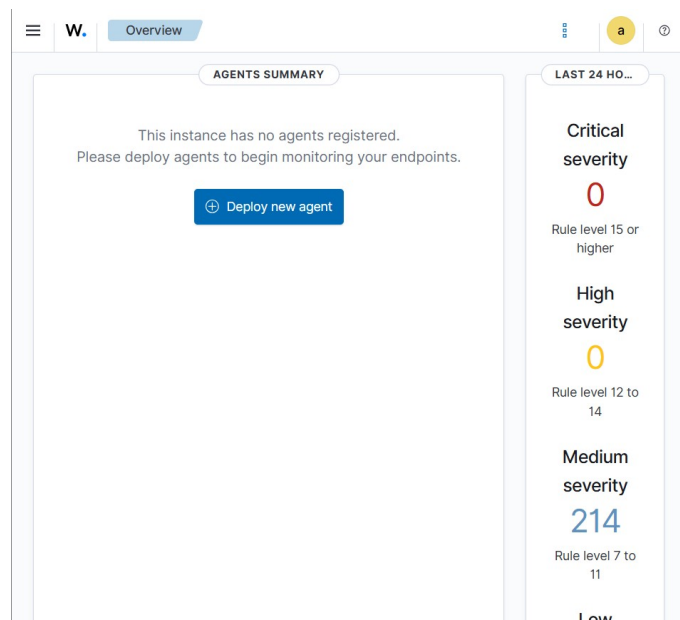
5. Wazuh Login Interface Verification

With the tunnel active, the Wazuh HTTPS login interface loaded successfully from the Windows browser. Reaching the login page confirmed that the Wazuh dashboard service was not only running at the operating-system level, but was also reachable through the configured access path from the host system.



6. Dashboard Authentication and Functional Access

The administrator account was used to authenticate to the Wazuh interface. After login, the Overview page loaded successfully and displayed the Agents Summary and Last 24 Hours Alerts sections. The dashboard reported that no agents were registered at this stage, which is expected because Day 18 verifies platform health and accessibility rather than endpoint-agent deployment. The presence of the Overview interface and alert summary confirmed functional application-level access to the Wazuh Dashboard.



7. Storage Check and Final Service Revalidation

A final server-state check was performed after the dashboard test. The root filesystem reported a size of 97 GB, with approximately 19 GB used and 74 GB available, leaving the filesystem at 21% utilization. A combined systemctl is-active query then returned active for the Wazuh manager, indexer, dashboard, and Filebeat. This final check confirmed that sufficient storage remained available and that the four required services were still running after browser access had been validated.

```

matthew@trioscyber:~$ df -h /
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/ubuntu--vg-ubuntu--lv  97G   19G   74G   21% /
matthew@trioscyber:~$ sudo systemctl is-active wazuh-manager wazuh-indexer w
azuh-dashboard filebeat
active
active
active
active
matthew@trioscyber:~$

```

8. Installation and Post-Installation Troubleshooting Record

The Wazuh deployment required several corrective actions before the final health check could be completed. These actions are recorded below because the Day 18 task explicitly requires the installation troubleshooting to be documented. The corrective work was limited to the authorized laboratory virtual machine and its local VirtualBox configuration.

Issue observed	Evidence / symptom	Corrective action	Final result
Insufficient root filesystem capacity	The first installation attempt reached "No space left on device" while the VM had a 100 GB virtual disk but a much smaller root logical volume.	The LVM physical volume and root logical volume were expanded so the guest could use almost the full virtual disk.	Root filesystem increased to approximately 97 GB; Day 18 shows 74 GB available.
Incomplete / residual Wazuh package state	Failed installation attempts left Wazuh components partially installed and prevented a clean reinstall.	Residual wazuh-manager package state and /var/ossec content were removed, dpkg was repaired, and the installation was started again from a clean state.	The final all-in-one installation completed successfully.
Wazuh API initialization timing	The installer reported that the wazuh API user was not registered during the password-update stage.	The installation script was adjusted to allow additional API initialization time before requesting the API token, then the all-in-one workflow was rerun.	The installation progressed through dashboard initialization and finished successfully.
VirtualBox disk I/O interruption	VirtualBox paused the VM with BLKCACHE_IOERR during the installation process.	Host storage availability was checked, the VM was resumed, and the installation process was allowed to continue.	The installation completed without rebuilding the virtual machine.
Filesystem inconsistency after I/O interruption	A later boot entered initramfs and reported an unexpected ext4 filesystem inconsistency.	The root logical volume was repaired manually with fsck.ext4 and checked again until the second pass completed cleanly.	Ubuntu booted normally and the final Wazuh component check returned all four services active.

9. Wazuh Health Check Summary

The evidence collected during Day 18 was consolidated into a final operational summary. The service-level checks and application-level browser checks agree: the Wazuh all-in-one server was reachable and its core components were operational after troubleshooting and filesystem recovery.

Check	Observed evidence	Result
Remote server access	SSH session to trioscyber established through localhost:2222	Successful
Wazuh manager	systemctl is-active returned active	Operational
Wazuh indexer	systemctl is-active returned active	Operational
Wazuh dashboard	systemctl is-active returned active	Operational
Filebeat	systemctl is-active returned active	Operational
Dashboard login page	HTTPS login interface loaded through localhost:8443	Accessible
Dashboard Overview	Authenticated Wazuh Overview page displayed	Accessible
Root storage	97 GB root filesystem, 74 GB available, 21% used	Healthy

10. Conclusions

The Wazuh Component Health Check was completed successfully after the all-in-one installation. The Wazuh manager, indexer, dashboard, and Filebeat services were verified as active with systemctl, the HTTPS login interface was reachable through the SSH tunnel, and successful authentication opened the Wazuh Overview page. The final storage check also confirmed that the expanded root filesystem had adequate free space. The installation troubleshooting—including storage expansion, cleanup of incomplete package state, API initialization timing, VirtualBox I/O recovery, and filesystem repair—was documented and resulted in a stable, accessible Wazuh server suitable for the next laboratory activities.